Western Governor's University

D212 Task 1: Clustering Techniques

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8/12/2025

One of the most critical factors in patient relationship management that directly affects a hospital's long-term cost-effectiveness is understanding the patients and the conditions leading to hospital admissions. When a hospital understands its patients' characteristics, it is better able to target treatment to patients, resulting in a more effective cost of care for the hospital in the long term.

You are an analyst for a hospital that wants to better understand the characteristics of its patients. You have been asked to use clustering techniques to analyze patient data to identify groups of patients with similar characteristics, ultimately enabling better business and strategic decision-making for the hospital.

Part I: Research Question

A. Describe the purpose of your data mining report by doing the following:

1. Propose one question relevant to a real-world organizational situation that you will answer using one of the following clustering techniques:

Is it possible to use known patient characteristics to group patients using K means clustering with centroids?

2. Define one goal of the data analysis. Ensure your goal is reasonable within the scope of the selected scenario and is represented in the available data.

One goal of this data analysis is to build a useful grouping of clusters that hospitals can use to find an estimated range of initial days for some patients.

Part II: Technique Justification

B. Explain the reasons for your chosen clustering technique from part A1 by doing the following:

1. Explain how the clustering technique you chose analyzes the selected data set. Include expected outcomes.

K means clustering begins by randomly assigning each data point to a specified K number of clusters, creating centroids within each cluster, representing the mean measurements of that cluster. The K means model then assigns each observation to the nearest centroid, based on Euclidean distance. Upon each

observation grouping, both affected cluster's centroids are recalculated. This process repeats until all observations can no longer re-assign data points to closer groupings.

This analysis is expected to use continuous variables from the medical dataset to group patients into three to five centroids.

(Frost)

(GeeksforGeeks, "K Means Clustering – Introduction")

2. Summarize one assumption of the clustering technique.

One assumption of K means clustering is that cluster sizes will be equally distributed among chosen K clusters.

(GeeksforGeeks, "Demonstration of KMeans Assumptions")

3. List the packages or libraries you have chosen for Python or R, and justify how each item on the list supports the analysis.

Sklearn.cluster Kmeans – This will allow implementation of KMeans algorithm to determine the optimal cluster amount. (scikit-learn)

Pandas – Allows transformation of csv data into dataframe and manipulation of created dataframe.

Sklearn.preprocessing – Data scaling must be implemented to allow for clustering, this includes standardscaler which will do so.

Matplotlib.pyplot – Allows creation of scatterplot to visualize centroids and KMeans elbow method on a line chart.

Seaborn – Allows creation of scatterplot using dataset variables.

SKlearn.metrics Silhouette_score: This allows for a model silhouette score to evaluate the clusters.

Part III: Data Preparation

C. Perform data preparation for the chosen data set by doing the following:

1. Describe one data preprocessing goal relevant to the clustering technique from part A1.

The dataset will need to be standardized using standard scaler before running the K means algorithm.

(GeeksforGeeks, “Clustering in Machine Learning”)

2. Identify the initial data set variables you will use to perform the analysis for the clustering question from part A1, and label each as continuous or categorical.

The chosen dataset variables will be the continuous variables, ‘Income’, ‘VitD_levels’, and ‘Initial days’.

3. Explain each of the steps used to prepare the data for the analysis. Identify the code segment for each step.

Lines 9-11: First, the data will be loaded from the csv file into a pandas dataframe.

Lines 13-14: A new dataframe labeled ‘X’ will be defined to store only the columns which will be used in cluster analysis. This step will also round numeric variables.

Lines 16-18: The newly defined X dataframe is now scaled using standard scaler and the scaled dataframe is given a unique name.

Lines 24-29: KMeans algorithm is implemented to assess the optimal K number of clusters.

(GeeksforGeeks, “Elbow Method for Optimal Value of K in KMeans”)

(GeeksforGeeks, “K Means Clustering – Introduction”)

(Iván Palomares Carrascosa)

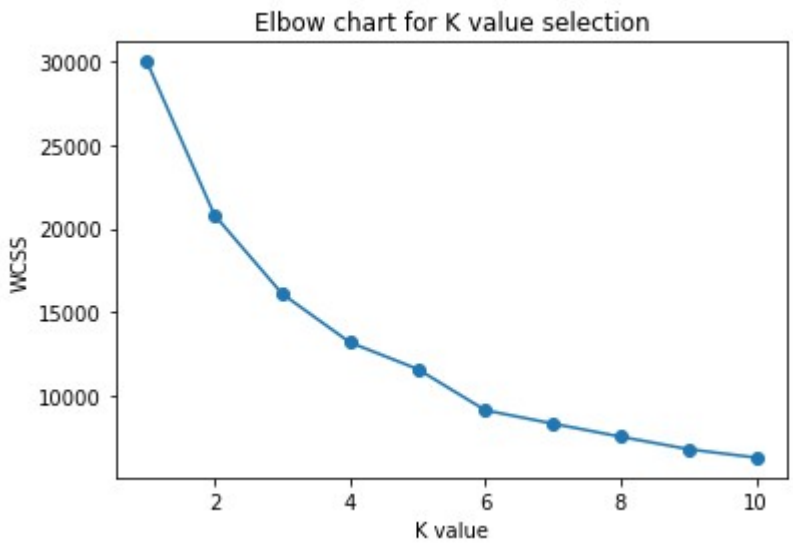
4. Provide a copy of the cleaned data set.

See file attached: “X_scaled.csv”

Part IV: Analysis

D. Perform the data analysis, and report on the results by doing the following:

1. Determine the optimal number of clusters in the data set, and describe the method used to determine this number.



The optimal K number of clusters was recommended by elbow method to be set at three. The implemented elbow method works by creating a visualization of the within cluster sum of squares at each possible K value in a selected range, allowing users to visually see the 'elbow', the point where the rate of change in WCSS takes a sharp decrease. This point indicates diminishing returns in increasing K values and can therefore be selected for cluster analysis.

(GeeksforGeeks, "Elbow Method for Optimal Value of K in KMeans")

(scikit-learn)

2. Provide the code used to perform the clustering analysis technique.

See file attached: "BrandonVelezD212Task1.py"

Part V: Data Summary and Implications

E. Summarize your data analysis by doing the following:

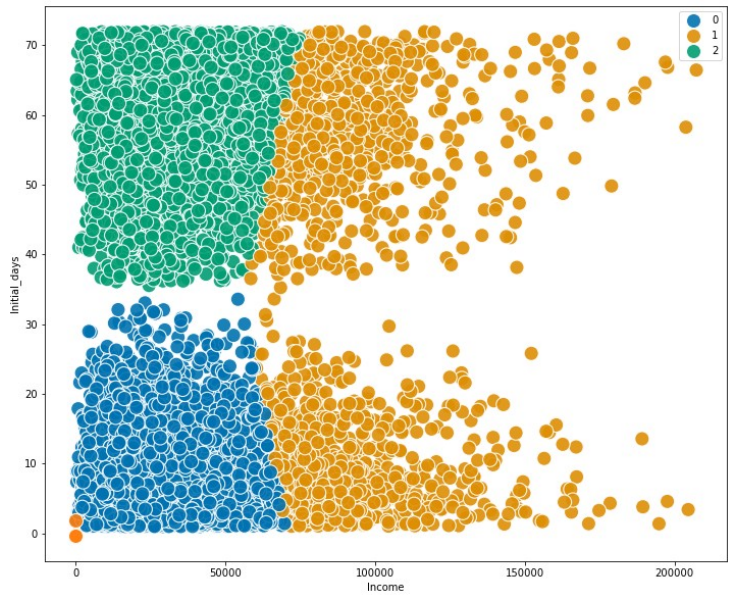
1. Explain the quality of the clusters created.

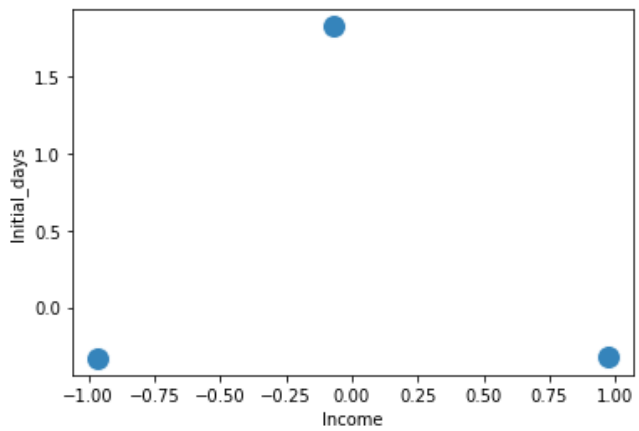
This analysis suggests that patients be classified into the following clusters:

Cluster 0: Patients with low income and low initial days.

Cluster 1: Patients with all ranges of income and average initial days.

Cluster 2: Patients with high income and low initial days.





2. Discuss the results and implications of your clustering analysis.

This clustering analysis has suggested that hospitals group patients into one of three clusters- low income & low initial days, the whole range of income & average initial days, and high income low initial days.

The silhouette score of 0.34 indicates that the clustering model is far from the perfect score of 1 which would indicate that clusters have no overlap and can always successfully predict which cluster patients would be grouped into, but there is enough validity to the model that it can be used to provide a rough prediction for initial days based on cluster.

(Chaudhary)

3. Discuss one limitation of your data analysis.

One limitation of this data analysis is that clustering is only implemented on numeric variables, not considering any of the information provided by categorical data.

4. Recommend a course of action for the real-world organizational situation from part A1 based on the results and implications discussed in part E2.

Based on the findings from this Kmeans clustering around centroids analysis, it would be recommended that hospitals apply this clustering model on future patient data during their stay to estimate the range of necessary initial days.

Part VI: Demonstration

F. Panopto link: https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=a06cf38d-a88a-4637-8c51-b336000ea3a2

G&H:

Works Cited

Chaudhary, Mukesh. "Silhouette Analysis in K-Means Clustering." *Medium*, 5 June 2020, medium.com/@cmukesh8688/silhouette-analysis-in-k-means-clustering-cefa9a7ad111.

Frost, Jim. "What Is K Means Clustering? With an Example." *Statistics by Jim*, 31 Aug. 2022, statisticsbyjim.com/basics/k-means-clustering/.

GeeksforGeeks. "Clustering in Machine Learning." *GeeksforGeeks*, 15 Jan. 2018, www.geeksforgeeks.org/machine-learning/clustering-in-machine-learning/#. Accessed 23 July 2025.

---. "Demonstration of KMeans Assumptions." *GeeksforGeeks*, 9 Dec. 2023, www.geeksforgeeks.org/machine-learning/demonstration-of-k-means-assumptions/. Accessed 24 July 2025.

---. “Elbow Method for Optimal Value of K in KMeans.” *GeeksforGeeks*, 6 June 2019, www.geeksforgeeks.org/machine-learning/elbow-method-for-optimal-value-of-k-in-kmeans/#. Accessed 29 July 2025.

---. “K Means Clustering – Introduction.” *GeeksforGeeks*, 2 May 2017, www.geeksforgeeks.org/machine-learning/k-means-clustering-introduction/.

Iván Palomares Carrascosa. “Performing Cluster Analysis in Python: A Step-By-Step Tutorial.” *Statology*, 27 Sept. 2024, www.statology.org/performing-cluster-analysis-in-python-a-step-by-step-tutorial/. Accessed 23 July 2025.

scikit-learn. “Sklearn.cluster.KMeans — Scikit-Learn 0.21.3 Documentation.” *Scikit-Learn.org*, 2019, scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html.